

Cynthia Shao

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EDUCATION

Cornell University | GPA: 3.89

Ithaca, NY

Bachelor of Science in Computer Science and Electrical and Computer Engineering

Aug. 2023 – May 2027

- Relevant Courses: Computer Architecture, Advanced Compilers, Operating Systems, High-Level Digital Design Automation, Embedded Systems, Analysis of Algorithms, Functional Programming, Object-Oriented Programming and Data Structures, Discrete Structures, Digital Logic and Computer Organization, Circuits, Wines

EXPERIENCE

Cornell Capra Lab, Cornell University

June 2025 – Present

Undergraduate Research Assistant mentored under Adrian Sampson, Rachit Nigam (MIT)

Cornell University

- Built a finite-state-machine compiler with syntax, parser, and IR for Calyx hardware accelerator infrastructure
- Optimized FSM inlining/splitting heuristics to balance control locality with performance in hardware pipelines
- Implemented unified static control pass to optimize downstream Verilog synthesis and FSM recognition

Zhang Research Group, Computer Systems Lab

May 2024 – Present

Undergraduate Research Assistant mentored under Zhiru Zhang

Cornell University

- Implemented sparsity support for Allo, a compiler for modular hardware accelerator programming built on MLIR
- Generated comprehensive Verilog dataset with novel prompting and fine-tuned LLAMA LLMs on Lambda Labs
- Achieved a top-10 placement in the LLM4HWDDesign Contest, innovating AI-driven hardware design

Cornell University Unmanned Air Systems (CUAir)

May 2024 – Present

Autopilot Software Engineer

Ithaca, NY

- Designed Bezier-spline-based navigation in C++, implementing cubic curve interpolation, reducing mission flight time by 15%, improving energy efficiency by 20%, and enhancing mission planning accuracy by 75% in tight turns
- Created custom MAVLink protocol extension for spline waypoints and managed Pixhawk memory allocation, optimizing waypoint density for smooth curved paths while respecting flight controller RAM limitations

College of Computing and Information Science, Cornell University

January 2026 – Present

Introduction to Compilers Teaching Assistant

Ithaca, NY

- Helped design and answer questions regarding homework and programming assignments, aiding in creating a lexer, parser, type checker, IR generator, x86 assembly generator, and various compiler optimizations from scratch.

Electrical and Computer Engineering Department, Cornell University

Jan. 2025 – Dec. 2025

Digital Logic and Computer Organization Teaching Assistant

Ithaca, NY

- Designed and led lab sessions for 100 students in digital logic and computer architecture, creating a hands-on programmable RISC-V microprocessor lab with performance analysis components on an FPGA

Jane Street

August 2023, August 2024, September 2025

Jane Street GPU Mode Hackathon

New York, NY

- Placed top 10 out of 50 teams, optimized PyTorch sequence models for H100s using batching and JIT compilation

Jane Street Wise

- Won first place in quantitative estimation competition, and learned market-making through Figgie trading games

PROJECTS

Allo-to-XLS Hardware Compiler & LLM Agent System

Nov. 2025 – Dec. 2025

- Integrated XLS toolchain and conducted Verilator cycle-accurate simulation, validating compiler-generated systolic arrays and GEMM implementations achieving 243.90 MHz on FreePDK 45nm
- Performed physical synthesis using Synopsys Design Compiler and Cadence Innovus, demonstrating synthesis equivalence between compiler and LLM-generated hardware with identical area, power, and latency

Bril Compiler Optimizer

Aug. 2025 – Dec. 2025

- Implemented and benchmarked compiler optimizations and passes on an educational Bril IR in Rust and Python.
- Including but not excluding: LVN, GVN (SSA generation), LLVM transformations, Dead Code Elimination, Conditional Constant Propagation, Liveness Analysis, Loop Invariant Code Motion, Generic Dataflow Solver

TECHNICAL SKILLS

Languages: C/C++, Rust, Python, Java, Verilog, SystemVerilog, Assembly, R, MATLAB, RISC-V, Bash, LaTeX

Tools/Frameworks: Linux, Git, LLVM, Lambda Labs, Tensorflow, FireSim, CIRCT, CUDA, Docker, React, Remix, NodeJS, NGinx, PM2, PostgreSQL, Prisma, Conda, CAN, I2C, SPI

Hardware: HLS, Icarus Verilog, Verilator, Modelsim Altera, RISC-V, Quartus, LT-Spice, Breadboarding, Soldering